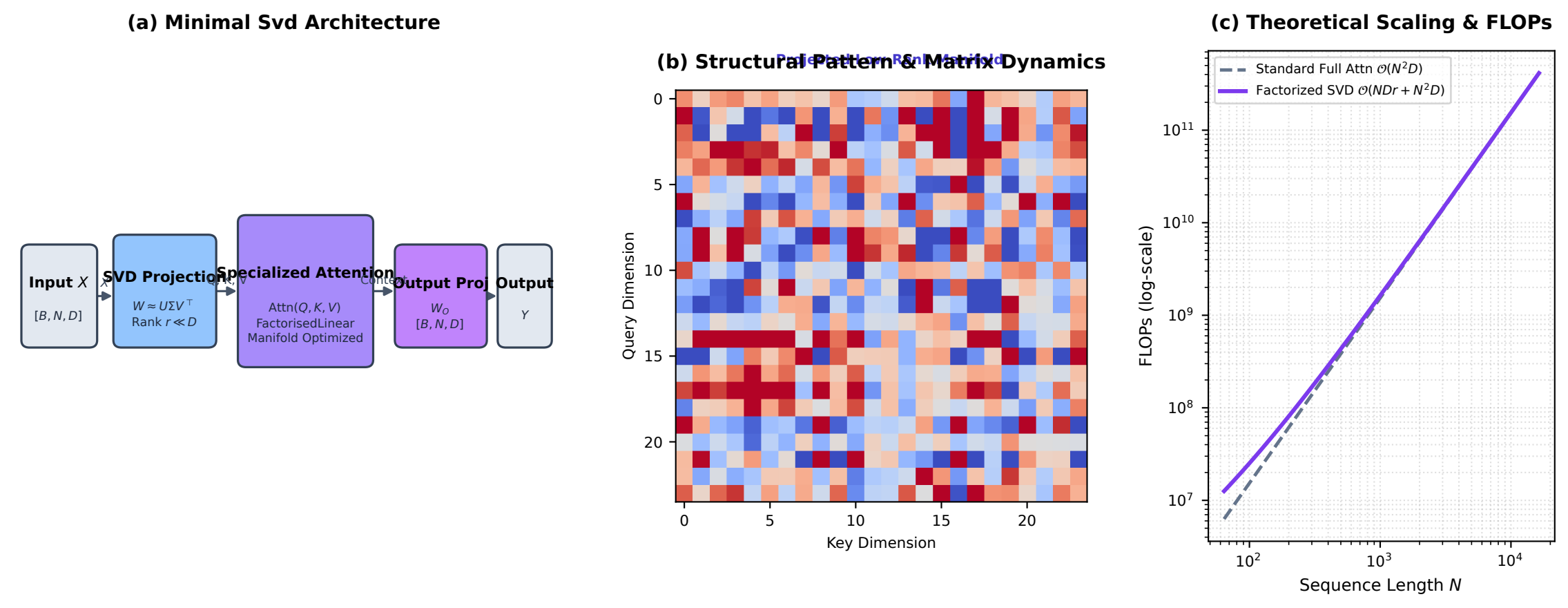
(a) Minimal Svd Architecture
Input $X$
$[B, N, D]$
$X$
SVD Projection
$W \approx U\Sigma V^\top$
Rank $r \ll D$
$Q, K, V$
Specialized Attention
Attn$(Q, K, V)$
FactorisedLinear
Manifold Optimized
Context
Output Proj
$W_O$
$[B, N, D]$
Output
$Y$
(b) Structural Pattern & Matrix Dynamics
Projected Low-Rank Manifold
Query Dimension
Key Dimension
0
5
10
15
20
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Factorized SVD $\mathcal{O}(NDr + N^2D)$
FLOPs (log-scale)
Sequence Length $N$
$10^7$
$10^8$
$10^9$
$10^{10}$
$10^{11}$
$10^2$
$10^3$
$10^4$